



M-Tech Production & Marketing

Technology | Training | Marketing | Projects

Week 4 - Day 5 Report

Internship Program — Batch 2026

Submitted By:	Ghazi Muhammad Abdullah – AI26025
Department:	AI / ML
Submitted To:	Malik Mohazin Zahid
Date:	24 July 2026

1. Introduction

Experiment No. 17 focused on Cloud AI Platforms, which have revolutionized how machine learning models are built, trained, and deployed by providing managed services that handle infrastructure, scaling, and operational complexity. Platforms such as Amazon SageMaker and Google Vertex AI offer end-to-end ML capabilities, eliminating the need for organizations to manage their own hardware, servers, and deployment pipelines. These platforms provide access to powerful GPUs for training, automatic model versioning and registry services, and one-click deployment to scalable inference endpoints. This experiment explored the core concepts, services, and workflows associated with cloud-based AI development on leading platforms.

2. Objectives

- Understand why cloud platforms are essential for modern AI/ML development.
 - Learn about the main managed AI services on AWS and Google Cloud.
 - Study the end-to-end cloud ML workflow from training to deployment.
 - Understand the concept of managed training jobs on GPU instances.
 - Explore model registries and inference endpoints.
 - Understand how cloud platforms handle scaling and operational management.
-

3. Task Description

Experiment No. 17 involved studying cloud AI platforms and their role in enterprise ML development. The complete training-to-deployment pipeline was examined, including dataset upload to cloud storage, launching managed training jobs, saving models to model registries, deploying to hosted endpoints, and scaling based on traffic demands.

Different cloud providers were compared, including AWS SageMaker, Google Vertex AI, and Azure Machine Learning. The strengths of each platform were studied, including their integration with ecosystem services, AutoML capabilities, and enterprise features. The concept of Hugging Face Inference Endpoints was also examined as a simpler alternative for hosting transformer models.

4. Work Done

Experiment No. 17 began with understanding why cloud platforms are preferred for AI/ML workloads compared to local development. Key advantages were identified, including access to powerful GPUs, scalable storage, managed infrastructure, global reach, and cost-effective on-demand resource usage.

The AWS SageMaker platform was studied in detail, exploring how data is uploaded to S3 storage, how training jobs are launched on managed instances, and how trained models are registered and versioned. Examples demonstrated the conceptual flow of submitting a training job and deploying the resulting model to a live endpoint.

Google Vertex AI was similarly examined, understanding its integration with Google Cloud storage, AutoML capabilities, and managed training infrastructure. The platform's strength in enterprise data integration and automated ML was explored.

The concept of inference endpoints was studied, understanding how deployed models are accessed over HTTPS, how they scale automatically with traffic, and how they handle prediction requests from client applications.

5. Challenges Faced

One of the main challenges in Experiment No. 17 was understanding the differences between cloud providers and their respective managed services. Each platform uses different terminology, has different integration points, and offers different capabilities. Developing a comparative understanding required careful study of each platform's offerings and strengths.

Another challenge involved understanding the complete cloud ML pipeline and how components such as data storage, training infrastructure, model registry, and deployment endpoints work together. This was addressed by studying the end-to-end flow and each component's role in the overall architecture.

6. Key Learnings

Experiment No. 17 provided valuable knowledge of cloud AI platforms and their importance in enterprise ML development. I learned how managed services handle the operational complexity of training and deploying models, allowing data scientists and ML engineers to focus on model development rather than infrastructure management.

The concept of on-demand GPU access was particularly insightful, understanding how cloud platforms provide access to powerful hardware that would be prohibitively expensive for most organizations to own and maintain. I also learned about model registries and how they enable versioning, collaboration, and reproducibility in ML workflows.

Understanding inference endpoints and automatic scaling gave me insight into how deployed models serve requests reliably, even under variable traffic conditions. The comparison between different cloud providers also helped me understand how platform selection depends on factors such as ecosystem integration, cost, and specific capabilities.

7. Conclusion

Experiment No. 17 successfully demonstrated the power and convenience of managed ML services for developing and deploying AI applications. By understanding how platforms such as AWS SageMaker and Google Vertex AI handle training, versioning, deployment, and scaling, I gained a solid foundation for leveraging cloud infrastructure in real-world AI projects.

Cloud platforms represent the future of AI development, enabling organizations to build and deploy sophisticated models without managing complex infrastructure. The concepts learned are essential for any ML engineer working in production environments.

Intern Signature:



Supervisor:

Name:

Ghazi Muhammad Abdullah

Name:

Date:

24 July 2026

Designation: